H.W # 04

M. Abdullah Siddiqui 23I-0617 F

Trace & Det Plane

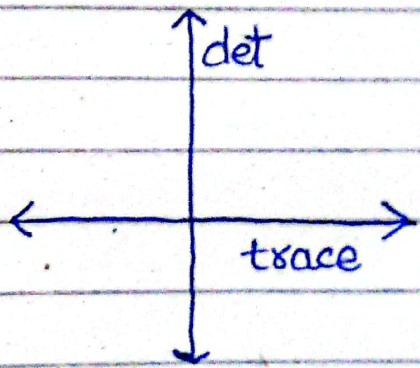
$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{trace}(A) = a + d$$
$$\det(A) = ad - bc$$

trace(A) provides the information about average rate of change of system.

The scalar on the other hand that is obtained from det(A) is the scaling factor of A.

trace is represented on X-Axis & determinant on Y-Axis on a 2D-plane.



Stability Analysis

Based on the plane, system's stability can be analysed:

Stable : $tr(A) < 0$ && $det(A) > 0$

Unstable: $tr(A) > 0$ && $det(A) > 0$

Saddle Point: $det(A) < 0$

Center : $tr(A) = 0$ && $det(A) > 0$